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Research paper

Graph-enhanced fault hierarchical perception for Dissolved Gas Analysis in power transformer

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ABSTRACT

Dissolved Gas Analysis (DGA) is a widely used technique for diagnosing power transformer faults by monitoring the gas content in transformer oil to identify potential issues. However, existing research often treats dissolved gas samples as loosely connected independent entities, failing to explicitly model the relationships between them. Moreover, the similarity between faults caused by different factors can sometimes be higher than that between faults caused by the same factor, highlighting the need for hierarchical modeling of fault relationships to improve diagnostic performance. To address these challenges, we propose a graph-enhanced fault hierarchical perception model for DGA. In this work, a K-Nearest Neighbors (KNN) graph is constructed by calculating the similarity between DGA samples in the transformer oil to capture potential associations between samples. To further enhance the learning of sample representations, we generate two augmented views by randomly perturbing node features and edges to capture information about different local structures in the graph. Additionally, we introduce a consistency constraint to ensure that the prediction results from the two augmented views remain consistent. Furthermore, we design a hierarchical perception-based ranking strategy that ranks fault similarities at a fine-grained level, effectively utilizing hierarchical fault information to optimize the model. Extensive experimental results demonstrate that the proposed model achieves high accuracy and strong generalization performance in fault-type prediction.

1. Introduction

Electric power transformers play an indispensable role in maintaining the stable operation of power systems. However, various factors such as improper operation and extreme weather conditions may lead to transformer failures. Failure to detect and diagnose transformer faults in a timely manner may result in transformer downtime, causing significant economic losses. Therefore, timely fault diagnosis of transformers is particularly important. Transformer fault diagnosis (TFD) (Abbasi, 2022; Rashed et al., 2025) refers to the monitoring, analysis, and identification of abnormal states or faults that may occur during the operation of electric power transformers to determine whether there are any issues with the transformer. Reliable transformer fault diagnosis technology can promptly detect and locate faults.

Given the practical significance of transformer fault diagnosis in real-world applications, both academia and industry have jointly driven the emergence of various diagnostic methods (Yang et al., 2024; Shao et al., 2025; Mao et al., 2022). Among them, Dissolved Gas Analysis (Liao et al., 2024; Zhang et al., 2024; Shu et al., 2024) (DGA) technology has gradually become a research hotspot and a mainstream technique in this field due to its unique advantages. Since 2023, IEEE Transactions journals alone have published 51 research papers on transformer fault diagnosis based on DGA technology, highlighting its critical role and broad application prospects in the field of transformer fault diagnosis. By regularly collecting samples of the transformer's insulation oil, DGA analyzes the concentration of dissolved gases to obtain crucial fault indication information, detecting and identifying potential faults to prevent further damage or shutdown of the transformer. Specifically, DGA identifies potential faults by monitoring the gas content in the transformer's insulation oil. The types and concentrations of gases in the insulation oil can provide crucial clues to engineers to determine whether there are potential faults inside the transformer. For example, changes in the type and concentration of gases may occur due to the decomposition of insulation materials, partial discharge in the oil-paper insulation system, or internal overheating of the transformer. By quantitatively and qualitatively analyzing these

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gases, fault types such as discharge, overheating, or other abnormalities can be identified. These analytical results provide important evidence for timely maintenance and repair actions.

With the development of machine learning, existing works have combined DGA with various machine learning techniques such as Support Vector Machine (SVM) (Zhou et al., 2022a), K-Nearest Neighbors (KNN) (Patil et al., 2024), Decision Tree (DT) (Zhou et al., 2022b), and Back Propagation Neural Networks (BPNN) (Liu et al., 2022) to enhance the detection accuracy of the models. This is attributable to the ability of machine learning models to identify fault patterns and predict potential fault types in the future by learning from a large amount of historical DGA data. This integration provides power system maintenance personnel with more accurate and timely fault diagnosis services, contributing to the stability and reliability of the power system.

Despite significant progress in DGA research, two primary limitations persist. Firstly, **many existing studies inadequately model the interrelationships between DGA samples, thereby limiting the overall effectiveness of transformer fault diagnosis.** To overcome this, it is crucial to implement methods that can accurately capture the underlying correlations among samples to enhance diagnostic accuracy. Secondly, **current models often overlook the intricate hierarchical structure among fault types, leading to suboptimal diagnostic results.** The hierarchical relationships among fault categories provide valuable insights, and a comprehensive exploration and utilization of this information are essential for improving the performance of transformer fault diagnosis. Therefore, the development of diagnostic models that skillfully incorporate hierarchical fault information is of paramount importance.

To overcome the challenges previously outlined, we propose an innovative approach termed the **G**raph-enhanced **F**ault **H**ierarchical **P**erception Model for Dissolved Gas Analysis (GFHP). Our model is designed to capture the intricate relationships between samples and the hierarchical nature of faults, thereby improving the accuracy and effectiveness of transformer fault diagnosis. Precisely, GFHP comprises three principal components: **the Graph-enhanced Representation Module, the Fault Hierarchy Perception Module, and the Fault Detection Module.** The Graph-enhanced Representation Module transforms DGA data into graph and produces two enhanced graph views to aid in understanding the inter-sample relationships. We incorporate constraints to maintain coherence in the semantic and structural information learned from these perspectives; The Fault Hierarchy Perception Module organizes fault types into separate clusters, assesses the proximity of different faults, and employs a customized loss function for model refinement; The Fault Detection Module is tasked with identifying faults using the representations gleaned from the preceding modules. Through the integration of these three components, the GFHP model strives to improve DGA fault diagnosis by effectively encapsulating sample relationships, harnessing hierarchical fault data, and precisely pinpointing faults.

The contributions of our work can be summarized as follows:

- **Novelty:** We propose a graph-enhanced fault hierarchy perception model, which aims to more accurately capture the complex relationships between faults in DGA data to improve transformer fault detection performance. To the best of our knowledge, this is the first work to reveal the hierarchical relationships among DGA fault categories and effectively apply them to fault diagnosis.
- **Technology:** We innovatively adopt feature masking and edge dropout strategies to generate diverse augmented views, thereby guiding the model to learn key features and global structural information. Moreover, based on the hierarchical relationships among faults, we carefully design a fault category contrastive strategy, which effectively guides the model in understanding and learning these hierarchical relationships.
- **Experimental Superiority:** Through comprehensive experiments carried out on two publicly accessible DGA datasets, we showcase the superior performance of our proposed approach, underscoring its effectiveness and generalizability.

The structure of the remainder of this paper is as follows: In Section 2, we present the definition of the research problem, validate the effectiveness of Graph Neural Networks (GNNs) on DGA data, and demonstrate the hierarchical fault relationships inherent in DGA data via data analysis. In Section 3, we provide a comprehensive overview of our methodology and explain the intricate details of our framework. The experimental results obtained from our research are analyzed in Section 4. Finally, in Section 5, we provide a concise summary of our work and contributions.

2. Preliminaries

2.1. Notations and problem statement

Let $\mathbf{X} \in R^{n\times d}$ denote the original feature matrix containing n samples of DGA data, with each sample represented as a d dimensional vector $\mathbf{x}_i$. In our work, $\mathbf{X}$ represents the matrix of the concentrations of five dissolved gases collected from the DGA-2000 or DGA-200 dataset, including H2 (hydrogen), CH4 (methane), C2H6 (ethane), C2H4 (ethylene), and C2H2 (acetylene). Therefore, n denotes the number of samples collected from the DGA-2000 or DGA-200 dataset(details are summarized in Section 4 Experimental Setup). Here, $d = 5$ corresponds to the above five gas concentrations. Based on the feature matrix, we construct a similar KNN graph denoted as $G = (V, \mathbf{X}, \mathbf{A})$. $V = \{v_1, v_2, \ldots, v_n\}$ is the set of nodes, where v_i represents the ith sample in the dataset. $\mathbf{A} \in R^{n\times n}$ represents the weighted adjacency matrix, where the value of each entry indicates the probability of connection between two samples. Perturbing G yields two augmented views, $G_1 = (V, \mathbf{X}_1, \mathbf{A}_1)$ and $G_2 = (V, \mathbf{X}_2, \mathbf{A}_2)$, where $\mathbf{A}_1$ and $\mathbf{A}_2$, $\mathbf{X}_1$ and $\mathbf{X}_2$ represent the weighted adjacency matrices and feature matrices of the two augmented views, respectively. $\mathbf{H}_1$ and $\mathbf{H}_2$ denote the node embedding matrices of G_1 and G_2. $\mathbf{H}$ is the final node representation matrix, and $\hat{\mathbf{y}}_i$ represents the model's predicted probability that sample i belongs to each fault category.

In the context of DGA, we formulate the transformer fault detection problem as a semi-supervised graph node classification task. The entire graph consists of two types of nodes, denoted as V_{train} and V_{test}. Specifically, V_{train} contains labeled nodes with the label set $\mathbf{Y}^L$, while the label set $\mathbf{Y}^U$ of V_{test} is inaccessible during training. We utilize a GNN as the prediction model $\mathcal{F}$, aiming to minimize the prediction error for the true labels $\mathbf{Y}^U$ of the unseen nodes in V_{test}.

2.2. Preliminary study

Graph Construction Effectiveness Analysis. To validate the effectiveness of graph construction, we have employed classical GNNs (Wang et al., 2023; Zhang et al., 2023; Gao et al., 2025) on the KNN graph constructed from the DGA-200 and DGA-2000 dataset. Representative machine learning methods such as BPNN, SVM, KNN, and DT have been selected as benchmarks. Table 1 summarizes the average and standard deviation of AUC-ROC and F1 score over 10 random splits of the test set.

The analysis of Table 1 highlights that GNNs exhibit a notable improvement in the AUC-ROC on both datasets, indicating that constructing graphs for DGA positively influences detection performance. However, the F1 score on both datasets suggests that there is still room for improvement. Previous studies, such as Shen et al. (2023), have demonstrated the effectiveness of graph augmentation techniques in enhancing representation learning capabilities. Besides, You et al. (2020) has highlighted that a combination of different types of augmentation strategies is more effective in improving node classification

Table 1
Performance Comparison of GNNs Based on Original Graph G with Classic Machine Learning Classification Methods (The best and second-best methods are highlighted in bold and underlined, respectively).

Methods		BPNN	SVM	KNN	DT	GNNs(w/G)
DGA-200	AUC-ROC	0.72 ± 0.07	0.75 ± 0.03	0.76 ± 0.03	0.77 ± 0.03	**0.88 ± 0.03**
	F1	0.63 ± 0.03	0.65 ± 0.05	0.68 ± 0.06	**0.70 ± 0.07**	0.67 ± 0.06
DGA-2000	AUC-ROC	0.79 ± 0.03	0.79 ± 0.01	0.83 ± 0.01	0.78 ± 0.02	**0.91 ± 0.01**
	F1	0.59 ± 0.03	0.73 ± 0.01	**0.79 ± 0.02**	0.73 ± 0.03	0.73 ± 0.02

performance than using a single strategy. Notably, You et al. (2020) has also pointed out that disrupting local structures and features, such as through edge dropping and feature masking, can effectively mitigate the risk of overfitting when sample sizes are small. Inspired by prior research, this paper further proposes a graph-augmented representation learning approach based on edge dropping and feature masking. Specifically, we apply random feature masking to the original graph to force the model to learn the invariance of key features, thereby reducing dependence on noisy features. At the same time, we adopt a random edge dropping strategy to disrupt the local graph structure, compelling the encoder to focus more on global semantic consistency rather than overfitting to specific neighborhood relationships.

Analysis of Fault Hierarchy Relation. The DGA-200 dataset encompasses seven distinct fault types: Low-Temperature Overheating (T1), Medium-Low Temperature Overheating (LMO), Medium Temperature Overheating (T2), High-Temperature Overheating (T3), Partial Discharge (PD), Spark Discharge (D1), and High-Energy Discharge (D2). Section 4 provides detailed statistical information regarding this dataset.

To investigate the relationships among different fault types, we construct a hierarchical tree based on the underlying physical causes of faults and conventional fault classifications, as illustrated in Fig. 1a. Distinct colors represent different causes (e.g., red for overheating and blue for discharge), while gradient colors within the same color differentiate various fault types under the same category. These faults are referred to be related. Intuitively, faults originating from the same cause (e.g., T1 and T2) exhibit greater feature congruence, whereas cross-cause faults (e.g., T1 and PD) demonstrate lower similarity.

To validate the hypothesis, we calculate the average cosine similarity of the samples for each fault category in the DGA-200 dataset and utilize this to quantify the similarity between fault types. The results are depicted in Fig. 1b. Upon scrutinizing Fig. 1b, we discern that cross-cause faults can exhibit higher similarity than faults within the same category. For instance, the similarity between T1 and PD surpasses that between T1 and T2, which contradicts our initial expectations. This finding implies the presence of noise features across different fault causes. Consequently, if the original graph is employed directly for representation learning without incorporating the hierarchical relationships among faults, the model may develop biases in capturing fault characteristics.

3. Methodology

In this section, we present the comprehensive workflow of the GFHP framework, illustrated in Fig. 2. To address the challenges posed by unmodeled relationships among samples and the underutilization of hierarchical information, we have devised three key modules to enhance the model's performance. **(1). Graph-enhanced Representation Learning Module:** This module initiates the process by constructing the initial graph from DGA samples. By explicitly modeling the inter-sample relationships, it captures information regarding various local graph structures through augmentation. **(2). Fault Hierarchy Perception Module:** Following the graph construction, this module refines node representations by leveraging insights extracted from the hierarchical tree. This step enhances the model's understanding of the hierarchical relationships among fault categories. **(3). Fault Diagnosis Module:** This module is tasked with classifying the refined representations, thereby enabling accurate fault diagnosis based on the enriched node representations. We provide a detailed description of the execution process of GFHP in Algorithm 1 and present the model architecture of GFHP on the DGA-200 dataset in Table 2.

3.1. Graph-enhanced representation learning

In this subsection, we describe the process of converting DGA data into a graph for graph representation learning. Considering the limited number of samples in the DGA dataset, we generate two augmented views by randomly perturbing the edges and node features of the initial graph. This approach captures different local structural information for the same samples, allowing the model to learn the relationships between samples from multiple perspectives. Besides, this strategy focuses on learning key features, thereby reducing dependence on noisy features. To fully exploit these augmented views, we use two Graph Convolutional Network (GCN) encoders with independent parameters to perform representation learning on each view.

Graph Construction. Given the feature matrix $\mathbf{X}$ of DGA samples, we expect to construct a graph where each sample is represented as a node, an edge is established between two nodes if their corresponding samples exhibit high similarity. Specifically, we construct the G based on cosine similarity by connecting each sample to its top-K most similar samples, as follows:

$$\mathbf{A}_{i,j} = \frac{\mathbf{x}_i^T \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \left\|\mathbf{x}_j\right\|}, \tag{1}$$

where $\mathbf{A}_{i,j}$ denotes the weight of the edge connecting sample i and sample j, $\mathbf{x}_i^T \cdot \mathbf{x}_j$ denotes the inner product of vectors in the ith and jth rows of matrix $\mathbf{X}$, $\|\cdot\|$ represents the norm of the vector.

To mitigate the performance constraints of the original graph stemming from an insufficient sample size, we incorporate graph augmentation techniques as proposed by Ding et al. (2022) and Zhao et al. (2021a). These techniques, which include random edge dropping and feature masking, are designed to preserve the global structural semantic consistency of the graph while highlighting the learning of pivotal features. This approach effectively boosts the performance of GNNs. The specific implementation details are as follows:

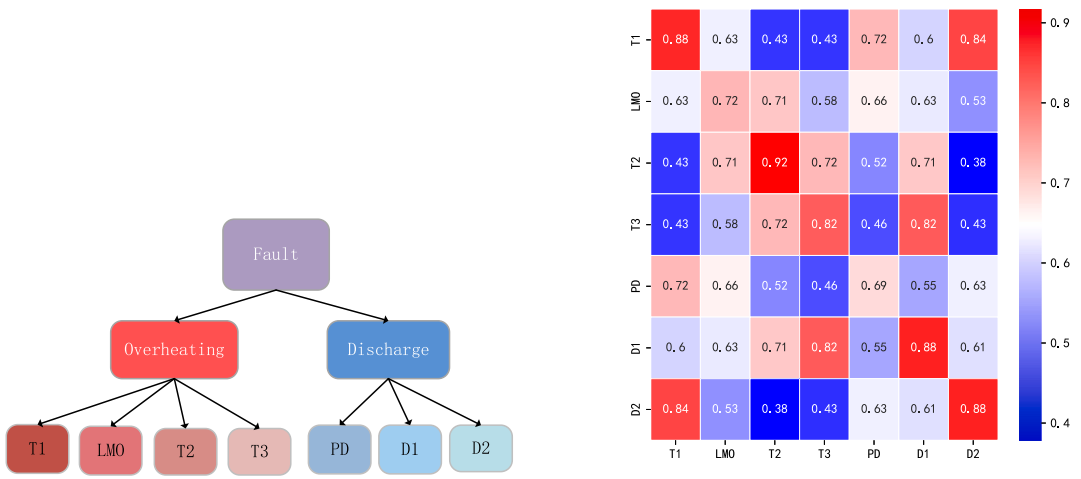
$$\begin{aligned} \mathbf{A_1}, \mathbf{A_2} &= DropEdge(\mathbf{A}) \\ \mathbf{X_1}, \mathbf{X_2} &= MaskFeat(\mathbf{X}), \end{aligned} \tag{2}$$

where $DropEdge(\mathbf{A}) = \{\mathbf{m}_1 \odot \mathbf{a}_1, \ldots, \mathbf{m}_1 \odot \mathbf{a}_n\}$, $MaskFeat(\mathbf{X}) = \{\mathbf{m}_2 \odot \mathbf{x}_1, \ldots, \mathbf{m}_2 \odot \mathbf{x}_n\}$, $\mathbf{a}_i$ and $\mathbf{x}_i$ denote the ith row of matrices $\mathbf{A}$ and $\mathbf{X}$, respectively. $\odot$ denotes the element-wise multiplication. $\mathbf{m}_1, \mathbf{m}_2 \in \{0,1\}^d$ represent the mask vectors, where the probability of each value being 0 is determined by a Bernoulli distribution, i.e., $\mathbf{m}_{ij} \sim Bernoulli(p)$. For instance, considering the jth value of $\mathbf{m}_1$, the probability of $\mathbf{m}_{1,j} = 1$ is p, and $\mathbf{m}_{1,j} = 0$ is 1-p.

Enhanced Graph Representation Learning. After constructing the transformer fault samples into two augmented graphs, G_1 and G_2, we use two GCN encoders with independent parameters to perform representation learning on each view, thereby obtaining their unique and rich feature representations, $\mathbf{H}_1$ and $\mathbf{H}_2$ as follows:

$$\mathbf{H}_t = \sigma\left(\tilde{\mathbf{A}}_t ReLU\left(\tilde{\mathbf{A}}_t \mathbf{X}_t \mathbf{W}_t^{(0)}\right) \mathbf{W}_t^{(1)}\right), \tag{3}$$

where $\mathbf{H}_t \in R^{n \times f}$ is the learned representation matrix, and $\tilde{\mathbf{A}}_t = \tilde{\mathbf{D}}_t^{-\frac{1}{2}} \left(\mathbf{A}_t + \mathbf{I}\right) \tilde{\mathbf{D}}_t^{-\frac{1}{2}} \in R^{n \times n}$ is the symmetrically normalized adjacency



(a) Fault hierarchy tree based on DGA-200 (b) Fault category similarity heatmap on DGA-200

Fig. 1. Fault hierarchical relation analysis.

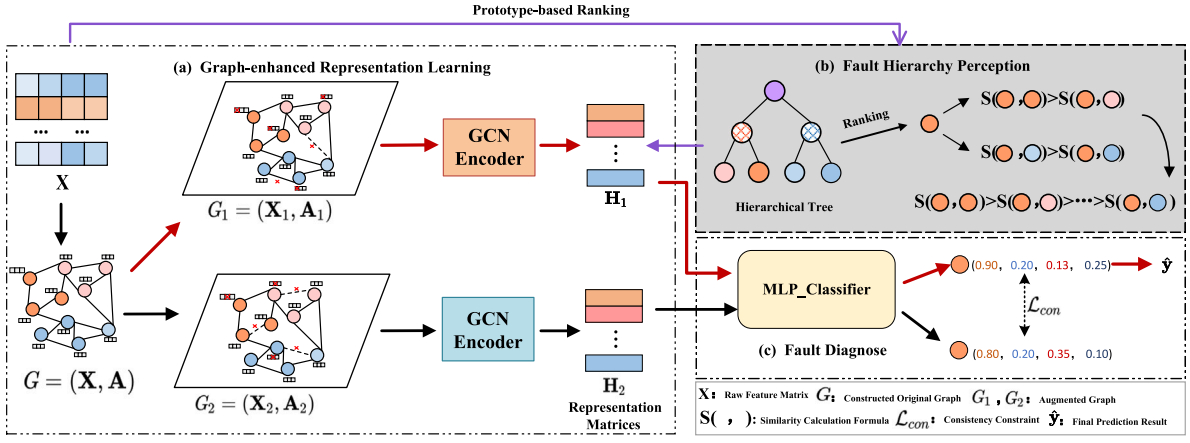


Fig. 2. The proposed GFHP framework. (a) Graph-enhanced representation module: Construct the original graph G based on the DGA feature matrix $\mathbf{X}$, enhance it to generate G_1 and G_2, and obtain the node representation matrices $\mathbf{H}_1$ and $\mathbf{H}_2$ through a graph encoder. (b) Incorporate the hierarchical information from the hierarchy tree into $\mathbf{H}_1$ to enhance its expressive power. (c) This module employs a Multi-Layer Perceptron (MLP) classifier to categorize the dual-view node representations and apply a consistency loss to constrain the prediction results.

matrix of the weighted undirected graph G_t with self-loops. $\mathbf{I} \in R^{n\times n}$ is the identity matrix, and $\mathbf{D}_{t_{ii}} = \sum_j (\mathbf{A}_t + \mathbf{I})_{ij} \in R^{n\times n}$ is the degree matrix. $\mathbf{W}_t^{(0)} \in R^{d\times h}$ and $\mathbf{W}_t^{(1)} \in R^{d\times f}$ are the trainable weight matrices for the specific layers. $t \in \{1,2\}$ denotes the different augmented views. $\sigma(\cdot)$ represents the activation function, such as softmax$(\cdot)$.

Since the fault hierarchy perception module employs loss based on class prototypes (i.e., class centers) to constrain the samples, this can lead to over-smoothing during parameter updates in gradient backpropagation. Specifically, samples from the same class may be pulled too close to the class center, diminishing intra-class differences. To mitigate this issue, we use $\mathbf{H}_2$ exclusively to enforce consistency constraints on $\mathbf{H}_1$, thereby preserving the essential differences between samples of the same class while effectively capturing global structural and key feature information. Ultimately, the optimized $\mathbf{H}_1$, after applying the constraints, serves as the final node representation and is fed into the fault diagnosis module.

3.2. Fault hierarchy perception

Based on the analysis in Section 2.2, we observe that DGA faults exhibit a hierarchical structure, where fault categories are not entirely independent but are instead interconnected through certain hierarchical relationships. However, existing methods have not fully leveraged this hierarchical relationship for DGA fault detection, thus limiting model performance. In fact, this hierarchical structure contains valuable prior information that can be used to enhance fault detection performance. To address this, we propose the Fault Hierarchy Perception module to strengthen the representation learning process. Next, we meticulously design a contrastive loss strategy aimed at leveraging the hierarchical relationships between faults to impose explicit constraints on the relationships among different fault categories. Then, through the backpropagation mechanism, the encoder is guided to learn these hierarchical relationships, ensuring that the model effectively integrates hierarchical information among fault categories, thereby enhancing detection performance.

The hierarchical relationships in the constructed hierarchy tree (as shown in Fig. 1a) can be described as follows:

(1) Taking faults caused by overheating as an example, the fault categories T1, LMO, T2, and T3 form **closely related hierarchical relationships**. This is because these faults share the same underlying cause, which is high temperature.

(2) The relationship between fault categories caused by overheating and those caused by discharge is considered a **loosely related hierarchical relationship**.

Table 2
Detailed Structure of GFHP for DGA-200.

Graph	Module	Input size	Layer type	Output size
View 1	GCN Encoder	(201,5)	GCNConv(5,128)	(201, 128)
		(201, 128)	ReLU	(201, 128)
		(201, 128)	GCNConv(128,64)	(201, 64)
		(201, 64)	Softmax	(201, 64)
	MLP_Classifier	(201, 64)	Linear(64, 7)	(201, 7)
		(201, 7)	Batch Normalization	(201, 7)
View 2	GCN Encoder	(201,5)	GCNConv(5,128)	(201, 128)
		(201, 128)	ReLU	(201, 128)
		(201, 128)	GCNConv(128,64)	(201, 64)
		(201, 64)	Softmax	(201, 64)
	MLP_Classifier	(201, 64)	Linear(64, 7)	(201, 7)
		(201, 7)	Batch Normalization	(201, 7)

Although the underlying causes of these two types of faults differ, they may still exhibit certain similarities or dependencies. When performing fault category hierarchy perception ranking, we prioritize the similarity between fault categories with closely related hierarchies over those with loosely related hierarchies. This means that fault categories within the same hierarchical group (e.g., T1, LMO, T2, T3) are expected to have a higher similarity compared to fault categories from different hierarchical groups. For example, considering overheating faults, their hierarchical relationship ranking may be as follows:

$$D\left(overheating, overheating\right) > D\left(overheating, discharge\right), \tag{4}$$

where $D\left(\cdot\right)$ represents the hierarchical relationships between fault inducements.

After determining the hierarchical relationships between fault categories, we perform fine-grained modeling of the hierarchical relationships between fault categories caused by the same underlying factor based on similarity. Specifically, we first calculate the cosine similarity between all pairs of samples using the original feature matrix $\mathbf{X}$ to measure the similarity between samples. Then, by utilizing their respective category labels, we compute the average similarity between samples within each category and between different categories, providing valuable insights into the similarity between fault categories. Based on the resulting average similarity, we rank the fault categories within the same hierarchical relationship. Finally, we stratify the fault categories according to these hierarchical relationships, prioritizing those with closer relationships. For example, in the DGA-200 dataset, the hierarchical ranking for overheating faults is as follows:

$$\begin{aligned} &S\left(\mathbf{X}_{T_1}, \mathbf{X}_{T_1}\right) > S\left(\mathbf{X}_{T_1}, \mathbf{X}_{LMO}\right) > S\left(\mathbf{X}_{T_1}, \mathbf{X}_{T_2}\right) > S\left(\mathbf{X}_{T_1}, \mathbf{X}_{T_3}\right) \\ &S\left(\mathbf{X}_{T_1}, \mathbf{X}_{D_2}\right) > S\left(\mathbf{X}_{T_1}, \mathbf{X}_{PD}\right) > S\left(\mathbf{X}_{T_1}, \mathbf{X}_{D_1}\right), \end{aligned} \tag{5}$$

where $S\left(\cdot,\cdot\right)$ represents the cosine similarity calculation formula between fault categories, and $\mathbf{X}_c$ represents the mean prototype of the original features of samples with category c and $c \in C = \left\{T_1, T_2, \ldots\right\}$ represents a certain category. The normal category in the DGA-200 dataset is regarded as the least significant category in terms of hierarchical relationships compared to other fault categories. Therefore, it is positioned at the bottom among all fault categories. Moreover, when determining the relationship between the normal category and other fault categories, only cosine similarity values are taken into account.

Inspired by Ranking Info Noise Contrastive Estimation (RINCE) (Hoffmann et al., 2022; Ning et al., 2023), we design a loss strategy based on the constructed hierarchical relationships. The prototype of the fault category exhibiting the highest similarity within the closely related hierarchy serves as the positive sample for the target node. Simultaneously, we consider the prototypes of fault categories with lower proximity as negative samples. To quantify the loss, we employ the infoNCE loss, designated as l_i. Each category's prototype vector $\mathbf{h}_{proto-i}$ is calculated as the mean of samples within that category. Subsequently, we eliminate the current fault category prototype and proceed to the next fault category prototype. This process is repeated until only the prototype of the fault category with the lowest proximity remains. To be more precise, the loss function is defined as $\mathcal{L}_{rank} = \sum_{i=0}^{|C|} l_i$, where $|C|$ denotes the cardinality of the set of categories C.

$$l_i = -log \frac{exp\left(\frac{S\left(\mathbf{h}, \mathbf{h}'_{proto-i}\right)}{\tau_i}\right)}{\sum_{j \geq i}^{|C|} exp\left(\frac{S\left(\mathbf{h}, \mathbf{h}'_{proto-j}\right)}{\tau_j}\right)}, \tag{6}$$

The loss l_i represents the discrepancy between the target node $\mathbf{h}$ and the ith fault category, which is sorted based on hierarchical relationships and similarity. The function $S\left(\cdot,\cdot\right)$ denotes the cosine similarity between node representations. $\mathbf{h}'_{proto-i}$ and $\mathbf{h}'_{proto-j}$ represent the prototype vectors of fault categories, arranged in order of similarity distance from the target sample, where i and j indicate their respective ranks. $\mathbf{h}'_{proto-i}$ and $\mathbf{h}'_{proto-j}$ are the prototypes obtained by linearly mapping $\mathbf{h}_{proto-i}$ and $\mathbf{h}_{proto-j}$ to a feature space with the same dimensionality as $\mathbf{h}$. These vectors are arranged in the order of similarity distance from the target sample, where i and j indicate their respective ranks. The hyperparameter τ_i controls the temperature, with the constraint $\tau_i < \tau_{i+1}$.

This strategy guides GFHP in learning the hierarchical relationships between faults through backpropagation, ensuring that similarity gradually decreases as the hierarchical ranking of samples increases. Simultaneously, it employs similarity sorting to extract semantic information between fault categories sharing the same hierarchical relationship. By fully utilizing the hierarchical information among faults, GFHP enhances the accuracy of fault category identification.

3.3. Fault diagnosis module

Once the fault data features have been accurately modeled and feature representations with minimal noise information have been obtained, these learned representations are employed for transformer fault detection. For this purpose, the learned representations are connected to a multi-layer perceptron (MLP) for classification:

$$\hat{\mathbf{y}} = softmax\left(\mathbf{W}^{(3)}\mathbf{h} + \mathbf{b}\right), \tag{7}$$

where $\hat{\mathbf{y}} \in R^{|C|}$ with $|C|$ denoting the total number of fault categories, and $\hat{\mathbf{y}}$ represents the probability vector of the target sample belonging to each fault category. $\mathbf{W}^{(3)} \in R^{f \times C}$ is the transformation matrix for node representations, $\mathbf{b}$ is a bias term.

To ensure that the model effectively leverages global structures and learns key features while avoiding over-reliance on specific features or interference from noise, we impose a consistency constraint on two augmented views. Unlike the conventional approach (Huo et al., 2023), our constraint directly applies to the prediction results of the augmented views rather than at the representation level. This strategy helps the model gain a deeper understanding of the true data distribution, thereby enhancing its generalization ability.

Building on this, we further enforce the prediction results of the same nodes in augmented view 2 to align with those in augmented view 1. This is because augmented view 1, under the strict constraints of the ranking contrastive loss, already learns the hierarchical relationships among faults, making its predictions more accurate. Mathematically, the consistency constraint can be expressed as follows:

$$\mathcal{L}_{con} = D_{KL}\left(\hat{\mathbf{y}}^1 \parallel \hat{\mathbf{y}}^2\right), \tag{8}$$

where $\hat{\mathbf{y}}^1$ and $\hat{\mathbf{y}}^2$ represent the prediction results of the two augmented views, respectively. $D_{KL}(\cdot \parallel \cdot)$ denotes the calculation of the Kullback–Leibler divergence between two distributions.

The primary loss function of GFHP adopts the cross-entropy (Wu et al., 2022; Zhao et al., 2021b) loss, which is formalized as follows:

$$\mathcal{L}_{main} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{|C|}\mathbf{y}_{i,c}\log\hat{\mathbf{y}}_{i,c}, \tag{9}$$

where $\mathbf{y}_{i,c}$ represents whether the true label of the ith sample belongs to class c, where it is 1 if true and 0 otherwise. N denotes the total number of training samples. $\hat{\mathbf{y}}_{i,c}$ represents the final prediction probability of the ith sample belonging to class c.

Finally, the learning objective balances the importance of three optimization objectives as follows:

$$\mathcal{L} = \mathcal{L}_{main} + \lambda\mathcal{L}_{con} + \beta\mathcal{L}_{rank}, \tag{10}$$

where $\mathcal{L}_{main}$ denotes the main loss, $\mathcal{L}_{con}$ and $\mathcal{L}_{rank}$ represent the graph consistency loss and fault hierarchical loss, respectively. λ and β are hyperparameters used to control the balance between the losses, we set them to 0.4 and 0.6 in this paper.

To further clarify the implementation details of GFHP, we have selected the DGA-200 dataset as an example to illustrate the model architecture of augmented view 1 and augmented view 2, with the details presented in Table 2.

Algorithm 1 Algorithm of **GFHP**

Input: DGA samples $\mathbf{X}$, training labels $\mathbf{Y}^L$, number of classes C, hyperparameters λ, β , K, and number of training epochs T
Output: Prediction probability matrix of the unlabeled samples $\hat{\mathbf{Y}}^U$

Calculate the weighted adjacency matrix $\mathbf{A}$ of the DGA samples based on Eq. (1), construct the graph $G = (\mathbf{X}, \mathbf{A})$ for the DGA samples based on $\mathbf{A}$
Generate two augmented views $G_1 = (\mathbf{X}_1, \mathbf{A}_1)$ and $G_2 = (\mathbf{X}_2, \mathbf{A}_2)$ based on Eq. (2)
for $T \leftarrow 1, 2, \cdots$ **do**
    Acquire the node representation matrices $\mathbf{H}_1$, $\mathbf{H}_2$ for the augmented views G_1, G_2 via Eq. (3)
    Acquire the hierarchical relationship between fault categories based on the fault causes, as shown in Eq. (4)
    Obtain the fine-grained correlations between faults by calculating the cosine similarity between prototypes of fault categories at the same hierarchical level
    Compute the fault hierarchical loss $\mathcal{L}_{rank}$ according to Eq. (6)
    Generate the prediction probability matrix $\hat{\mathbf{y}}$ based on Eq. (7)
    Apply consistency constraints to two augmented views according to Eq. (8)
    Update the model by minimizing $\mathcal{L}_{main} + \lambda\mathcal{L}_{con} + \beta\mathcal{L}_{rank}$ with standard backpropagation
Predict the probability matrix $\hat{\mathbf{y}}^U$ of unlabeled data belonging to each fault class based on the trained model
return $\hat{\mathbf{Y}}^U$

4. Experiments

In this section, we conduct extensive experiments to assess the effectiveness of the proposed GFHP. Specifically, the experiments aim to address the following research questions:

- **RQ1:** How does the performance of GFHP compare to existing methods?
- **RQ2:** What is the contribution of each module in GFHP?
- **RQ3:** How interpretable is the GFHP model?
- **RQ4:** How do the main parameters in GFHP affect the model?
- **RQ5:** How effective is GFHP?

4.1. Experimental settings

Datasets. The experiments in this paper are performed on two benchmark datasets, namely DGA-200[1] and DGA-2000.[2] Detailed information about these datasets can be found in Table 3. It is worth mentioning that edges between nodes i and j are only computed once if they are mutual neighbors.

- **DGA-200**: This dataset consists of 201 DGA samples obtained from IEEE-Dataport, referred to as DGA-200 in this paper. It includes 7 fault categories: T1, LMO, T2, T3, PD, D1, and D2.
- **DGA-2000**: This dataset, obtained from GitHub, comprises 2321 transformer samples, collectively referred to as DGA-2000 in this paper. After removing duplicate samples, there are 1982 valid samples. Apart from the normal category (N), DGA-2000 includes 6 fault categories: T1, T2, T3, PD, D1, and D2.

Baselines. To evaluate the performance of the proposed GFHP model, we have selected three advanced methods for comparison: the improved tree-based model EBMT (Liao et al., 2024), which is specifically designed for DGA analysis, and two methods that innovatively adapt GNNs to the DGA domain—GCDGA (Zhang et al., 2024) and GraphSmin (Shu et al., 2024). Specifically, GCDGA explores the complex relationships in DGA data from both heterophilic and homophilic perspectives, while GraphSmin further advances this approach by expanding the sample size to enhance data diversity, thereby unlocking additional potential from the data.

Evaluation Metrics. To comprehensively evaluate the performance of GFHP, we utilize three commonly used metrics AUC-ROC, F1 score and Precision. All evaluation metrics have been computed using the corresponding functions from the Python library 'sklearn.metrics'. For the AUC metric, we have adopted the One-vs-Rest (OvR) classification strategy and averaged the results. Higher values of these metrics indicate better model performance.

Implementation Details. We implement GFHP using PyTorch and train GFHP using the Adam optimizer with a learning rate of 0.001. We mitigate overfitting by setting the dropout rate to 0.005. In the graph-enhanced module, we set the number of the convolutional layer and node hidden dimension to 2 and 128 for both DGA-200 and DGA-2000 dataset. In the experiment, we randomly split each dataset into 70% for training and 30% for testing. We conduct 10 rounds of experiments, with different data in each round while maintaining the same split ratio. The reported results are the mean and standard deviation of the test set. All baseline methods use the optimal hyperparameters reported in the respective papers. To ensure fairness in the experiments, all baseline methods and GFHP use the same training and testing datasets. Our code is available at https://github.com/mala-tang/GFHP.

4.2. Performance comparison (*rq1*)

In this subsection, we initially benchmark the performance of GFHP against all baseline approaches, subsequently delving into an analysis of its generalization across coarse-grained datasets.

Overall Performance Comparison. To address RQ1, we have conducted an evaluation to compare the performance of the baseline

[1] https://IEEE-dataport.org/documents/dissolved-gas-data-transformer-oil-fault-diagnosis-power-transformers-membership-degree
[2] https://github.com/alan-456/transformer-fault-dataset

Table 3
Statistical details of DGA datasets.

Dataset	Classes	Features	Nodes	Edges
DGA-200	7	5	201	586
DGA-2000	7	5	1982	5926

Table 4
Performance Comparison of GFHP with baselines (The optimal and suboptimal methods are highlighted in bold and underlined, respectively).

Method	DGA-200			DGA-2000		
	AUC-ROC	F1	Precision	AUC-ROC	F1	Precision
EMBT	0.88 ± 0.02	0.69 ± 0.05	0.71 ± 0.05	0.86 ± 0.02	0.74 ± 0.04	0.75 ± 0.04
GCDGA	0.89 ± 0.04	0.73 ± 0.05	0.72 ± 0.03	0.90 ± 0.01	0.76 ± 0.05	0.76 ± 0.02
GraphSmin	0.94 ± 0.08	0.73 ± 0.12	0.77 ± 0.03	0.94 ± 0.04	0.77 ± 0.10	0.76 ± 0.03
GFHP	**0.95 ± 0.02**	**0.82 ± 0.06**	**0.83 ± 0.03**	**0.96 ± 0.01**	**0.82 ± 0.01**	**0.82 ± 0.01**
GFHP-data	0.95 ± 0.03	0.87 ± 0.05	0.89 ± 0.04	0.90 ± 0.03	0.80 ± 0.03	0.80 ± 0.03
GFHP-aug	0.92 ± 0.04	0.80 ± 0.06	0.80 ± 0.04	0.93 ± 0.01	0.80 ± 0.02	0.80 ± 0.02
GFHP-oc	0.93 ± 0.04	0.81 ± 0.07	0.80 ± 0.05	0.94 ± 0.01	0.80 ± 0.02	0.80 ± 0.02

Table 5
Confusion Matrices and Precision of Fault Categories for DGA-2000 and DGA-200.

Datesets	DGA-2000							DGA-200						
	N	T1	T2	T3	PD	D1	D2	LMO	T1	T2	T3	PD	D1	D2
N/LMO	202	4	0	3	3	1	0	2	1	0	0	0	0	1
T1	6	17	1	6	5	0	0	0	7	0	0	2	0	0
T2	1	4	21	6	1	0	0	0	1	4	0	0	0	0
T3	2	1	2	126	0	1	0	0	0	1	12	0	0	0
PD	3	0	0	0	27	3	1	0	1	0	0	8	0	0
D1	4	1	1	4	1	64	8	0	0	0	0	0	13	0
D2	0	0	1	0	2	12	53	1	0	0	0	1	0	8
Precision	0.93	0.63	0.81	0.87	0.69	0.79	0.85	0. 67	0. 7	0.8	1	0.73	1	0.89

methods and our proposed GFHP method. The results of the comparison are presented in Table 4, and the following observations are made:

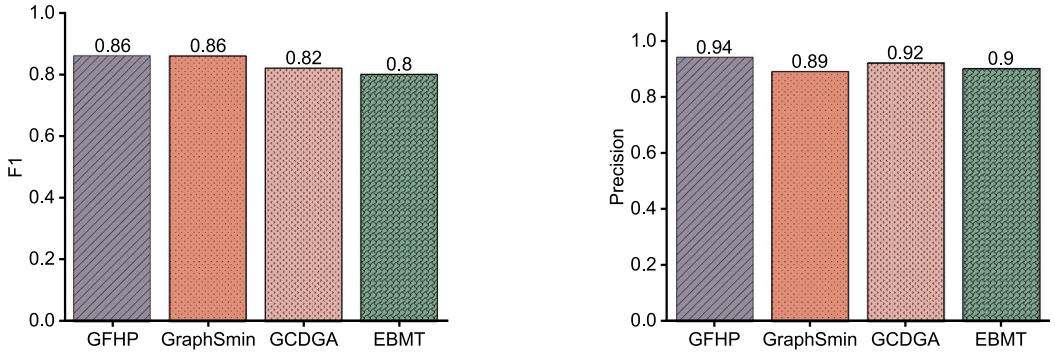
First, GFHP consistently outperforms all baseline methods across all datasets. This demonstrates the effectiveness of the graph augmentation strategy and the fault hierarchy perception module implemented in GFHP. The integration of fault hierarchy information into graph representations significantly contributes to its superior performance.

Second, GCDGA and GraphSmin are two advanced methods that utilize GNNs for transformer fault detection, revealing different types of relationships among DGA samples. Jointly analyzing with Table 1, we observe that these methods consistently outperform traditional machine learning approaches, including customized machine learning methods. We attribute this performance difference to the advantages of GNNs in capturing complex sample relationships and leveraging structural information. This further supports the effectiveness and feasibility of applying GNNs to transformer fault diagnosis.

Furthermore, to comprehensively evaluate the effectiveness of fault hierarchy perception in GFHP, we present the confusion matrices and the precision values for each class on the test sets of two datasets in Table 5. The value in row i, column j represents the number of samples with the true label i that are classified as label j. Higher values along the main diagonal indicate a greater number of correctly predicted samples. By analyzing Table 5, we find that GFHP, based on fault hierarchy perception, achieves better classification performance for most fault categories. Notably, GFHP effectively reduces feature confusion between faults caused by different physical factors. This is reflected in the fact that, among fault categories with relatively poor classification performance, misclassified samples are significantly less likely to be assigned to fault categories with different physical causes than to those within the same cause. However, fault hierarchy perception also increases the tolerance for feature overlap within the same physical cause to some extent. As a result, for certain categories with high feature overlap, such as T1, T2, and T3, as well as D1 and D2, there remains room for improvement in GFHP's performance. This provides a direction for future research.

Generalization Comparison. To further assess the model's generalization ability and evaluate its performance on coarse-grained data, we reconstruct the data based on the physical causes of fault categories in the DGA dataset and conduct additional experiments. Through this reconstruction, we generate two coarse-grained datasets: DGA-200-2 and DGA-2000-3. Specifically, DGA-200-2 is merged into two categories—overheating and discharge—based on physical causes, whereas DGA-2000-3 is divided into three categories—overheating, discharge, and normal—due to the presence of normal samples in the dataset. This coarse-grained representation allows us to evaluate the model's generalization performance on datasets with a reduced number of fault categories.

Fig. 3a presents the F1 scores of all methods on DGA-200-2, while Fig. 3b shows the Precision on DGA-2000-3. By analyzing Fig. 3, we observe that GFHP consistently achieves the best performance on both datasets. This suggests that even when the hierarchical relationships among faults are weakened, leaving only the relationships between different physical causes, GFHP can still capture the differences among these physical causes. Furthermore, by combining these findings with the experimental results in Table 4, we observe that all baseline methods exhibit improved performance on the coarse-grained datasets. This is because fault samples with the same physical cause share high feature similarity and are more likely to be misclassified into other fault categories. In contrast, fault patterns associated with different physical causes exhibit greater distinguishability. Consequently, merging faults with the same physical cause into a single category reduces classification difficulty, leading to performance improvements across all



(a) F1 on DGA-200-2 (b) Precision on DGA-2000-3

Fig. 3. Performance of GFHP and baselines on DGA-200-2 and DGA-2000-3.

methods. However, the fact that GFHP continues to outperform baseline methods further demonstrates its superior generalization ability across different datasets.

4.3. Ablation study (*rq2*)

To address RQ2, we have conducted ablation experiments from both the data and model perspectives. Three types of ablation experiments are performed, which are described below:

- **GFHP-data**: In the DGA-200 dataset, four fault categories related to overheating are selected, whereas in the DGA-2000 dataset, a larger number of discharge-related samples are chosen for retraining and testing the model.
- **GFHP-aug**: GFHP without the fault hierarchy perception module.
- **GFHP-oc**: GFHP without the graph consistency constraint.

(1) GFHP-data: On the DGA-200 dataset, GFHP-data exhibits a significant performance improvement, whereas its performance on the DGA-2000 dataset decreases substantially. By analyzing the data, we attribute the performance decline on DGA-2000 to the fact that failures induced by the same physical causes are more prone to misclassification, leading to reduced performance. In contrast, the anomalous performance improvement on DGA-200 is likely due to the smaller sample size of this dataset, which allows the model to more easily learn specific patterns in the data. (2) GFHP-aug: The performance decreases to varying degrees on both datasets, with a more pronounced decline compared to GFHP-oc. This phenomenon highlights the importance of capturing the hierarchical relationships between failures. (3) GFHP-oc: The F1 decreases on both datasets. This indicates that the graph consistency constraint helps the model learn and maintain consistency within the graph structure to some extent.

4.4. Visualization (*rq3*)

To further explore the capability of GFHP in learning hierarchical relationships between faults, we randomly select a subset of samples from all categories in the DGA-2000 test set for analysis. To intuitively present the sample distribution after learning with GFHP, we employ t-SNE visualization technique, with the results shown in Fig. 4.

Observing Fig. 4, it is evident that after learning with GFHP, the samples exhibit a significant clustering pattern, where points of the same color (representing samples from the same fault category) tend to cluster closely together. This indicates that GFHP can effectively capture the common features of similar faults. Further analysis has revealed that, except for the discharge-induced D1 fault, the inter-class distance among fault categories within the same hierarchical branch (e.g., T1, T2, and T3) is significantly smaller than that between

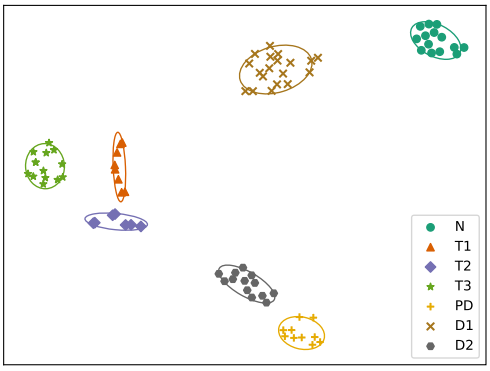


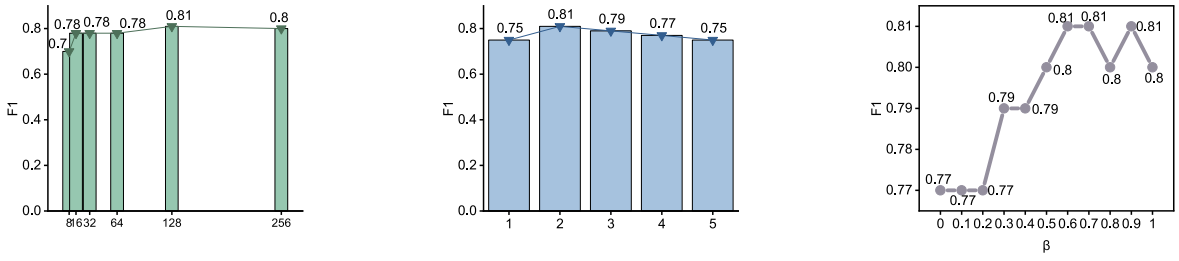
Fig. 4. Visualization of embedding spaces for randomly selected nodes from DGA-2000 dataset.

cross-hierarchy categories (e.g., T1 and PD). This result validates the effectiveness of GFHP in modeling the hierarchical structure of fault causes. The phenomenon where D1 maintains nearly equal distances from clusters of other faults may be attributed to its considerable feature overlap with other fault categories, which aligns with the tendency of D1 to be frequently misclassified during classification. These findings indicate that while GFHP can effectively learn and represent hierarchical relationships between faults, further research is needed to address fine-grained differentiation issues caused by feature overlap.

4.5. Hyper-parameter analysis (*rq4*)

In this section, we conduct an in-depth analysis of the impact of different settings of several key hyperparameters in the GFHP model on its performance. These hyperparameters include the hidden layer embedding dimension in the GNN, the number of GNN layers, and two critical hyperparameters, λ and β, which balance the ranking contrastive loss and the consistency loss. Since the sum of λ and β is fixed at 1 in this study, and given that the primary contribution of this work lies in the fault hierarchy-aware ranking contrastive loss, we focus on simplifying the analysis process and highlighting the research emphasis by presenting only the model performance as β varies.

(1) Impact of Hidden Layer Embedding Dimension in GNN: The embedding dimension of hidden layers plays a role in influencing the accuracy of feature learning in the graph representation module to some extent. By examining Fig. 5a, we can observe how the hidden layer embedding dimension affects the F1. It is evident that as the dimension of hidden layers increases, the F1 initially rises and then declines. The F1 reaches its highest point when the dimension is set to 128. However, surpassing this dimension threshold leads to overfitting,



(a) Embedding dimension (b) Number of hidden layers (c) Hyperparameter β

Fig. 5. The impact of hyperparameters on GFHP in DGA-200..

as excessive utilization of features causes the model to learn noise information from the samples, ultimately resulting in a decrease in model performance.

(2) Influence of the Number of Layers in GNN: Fig. 5b visualizes the impact of the number of convolutional layers stacked in the graph representation module on model performance. It is evident that the F1 score does not consistently increase as the number of layers increases. When there is only one convolutional layer, the model exhibits relatively poor accuracy and stability due to its limited capacity to capture complex relationships among samples. With two convolutional layers, the F1 score reaches its peak, indicating optimal performance. However, as the number of layers exceeds three, the model tends to overfit, resulting in a gradual decline in performance.

(3) Fig. 5c illustrates the variation in model performance as β changes. Since the sum of λ and β is fixed at 1, the variation in β indirectly reflects the impact of λ on model performance. The figure shows that when β is set to 0.6 or 0.7, the model achieves outstanding performance, with an F1 score reaching 0.81, and maintains relative stability within this range. This indicates that under these parameter settings, an optimal balance is achieved between the ranking contrastive loss and the consistency loss, enabling the model to maintain high accuracy while effectively addressing inconsistencies in the data. However, when β exceeds 0.7, the model performance exhibits noticeable fluctuations and a general downward trend. This phenomenon highlights the critical role of the consistency constraint in ensuring model stability. An excessively high β may cause the model to overly prioritize the ranking contrastive loss, thereby weakening its consideration of data consistency and ultimately impairing overall performance.

4.6. Effectiveness analysis (**rq5**)

To validate the effectiveness and practicality of GFHP, we have first conducted a theoretical analysis of its time complexity. Next, we have presented the spatiotemporal complexity of all methods on the DGA-200 dataset, with the corresponding experimental results shown in Fig. 6a. To further verify the practicality of GFHP, we have randomly selected samples of different proportions from the DGA-2000 dataset and recorded the corresponding changes in GFHP's time cost as the sample size gradually increases. The findings are depicted in Fig. 6b.

During the offline training phase, we have first constructed the adjacency matrix using cosine similarity, where the time complexity of computing cosine similarity between features is $O\left(n^2 d\right)$, where n represents the maximum number of nodes in the graph, and d is the dimension of the initial features. Based on the computed cosine similarity, we have then calculated the class prototypes for each category, which has a time complexity of $O\left(|C|\right)$ where $|C|$ denotes the number of categories. Therefore, the total time complexity of this preprocessing step is $O\left(n^2 d + |C|\right)$. Although this step incurs a relatively high computational cost, it is only required to be performed once in an offline manner, with its results being directly reusable in subsequent steps. Next, we have employed a graph encoder for representation learning, with a time complexity of $O\left(md + mh\right)$, where m denotes the number of nonzero elements in the adjacency matrix, and h represents the dimension of the hidden features. Moreover, the ranking contrastive loss based on class prototypes has a time complexity of $O\left(|C|\, n\right)$. Considering all the above steps, the total time complexity of the training process is approximately $O\left(md + mh + |C|\, n\right)$.

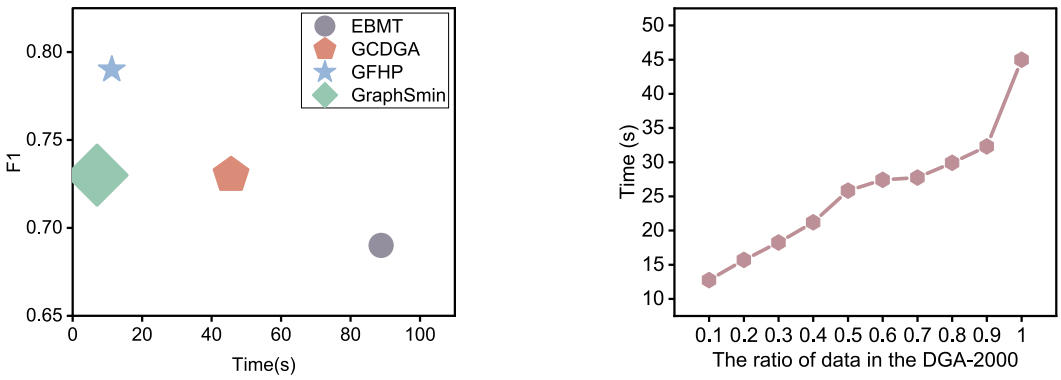
In Fig. 6(a), we have compared the spatiotemporal complexity and performance of GFHP with baseline methods on the DGA-200 dataset. In the figure, the x-axis represents training runtime, the y-axis represents the F1 score, and the shaded region reflects the relative number of parameters. GFHP has achieved high performance with relatively low spatiotemporal complexity, demonstrating the effectiveness of incorporating fault hierarchy relationships. In contrast, while GraphSmin has a lower runtime, it sacrifices a certain degree of spatial complexity.

Fig. 6(b) has detailed the trend of GFHP's runtime as the selected sample size from the DGA-2000 dataset increases. Notably, this growth has not followed a strictly exponential pattern but instead has exhibited a phased linear increase. Specifically, when the dataset size reaches certain thresholds, such as 50% and 100% of the DGA-2000 dataset, the growth in runtime becomes significantly larger. However, as the dataset size continues to increase, the runtime reverts to a linear growth pattern. This phenomenon suggests that GFHP possesses an adaptive adjustment capability when handling datasets of different scales, enabling it to efficiently accommodate increasing data volumes while maintaining computational efficiency. This phased linear growth pattern not only highlights the flexibility and robustness of GFHP but also provides a reliable estimation of runtime for large-scale fault detection tasks.

To further analyze the effectiveness of GFHP, we conduct a t-test comparing it with the strongest baseline, GraphSmin. The test results indicate that the p-value is 0.0820, which exceeds the commonly used significance level of 0.05. This suggests that, statistically, the performance difference between GFHP and GraphSmin is not significant. This result is expected, as GraphSmin represents the latest advancement in applying GNNs to the DGA domain and has already achieved state-of-the-art performance in this field.

5. Conclusion

This study focuses on exploring the hierarchical relationships among transformer fault categories. To achieve this, we introduce a novel method called Graph-enhanced Fault Hierarchy Perception (GFHP), specifically designed for analyzing the relationships between fault categories in dissolved gas analysis. The GFHP approach involves enriching the content and structure of transformer sample data and calculating various hierarchical distances between faults. Initially, we enhance the content and structure of the transformer sample data. Then, we propose a fault hierarchy perception ranking strategy that processes fault categories in a hierarchical manner, leveraging the rich information within each fault category. Through extensive experiments conducted on two widely used transformer fault detection datasets, we demonstrate that GFHP effectively captures the relationships between fault categories, resulting in improved outcomes for fault diagnosis.



(a) Spatiotemporal efficiency on DGA-200 (b) Amount of Data and Time Cost

Fig. 6. Spatiotemporal performance analysis of GFHP.

CRediT authorship contribution statement

Yingyue Zhang: Validation, Writing – original draft, Visualization, Methodology, Software, Formal analysis. **Huifang Ma:** Investigation, Supervision, Funding acquisition, Writing – review & editing, Conceptualization. **Yuwei Gao:** Investigation, Project administration. **Shengjiang Peng:** Supervision, Writing – review & editing, Conceptualization. **Ruijia Zhang:** Writing – review & editing, Conceptualization, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

I have shared the link to my data and code in the paper.

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